Knowledge Distillation and Deployment of a Lightweight Command-Parsing LM on a Local Cloud

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1. Architecture and design decisions

The project is a local cloud with four layers. A k3d cluster is created from `cluster/k3d.yaml` (one server, one agent, Traefik and metrics-server kept enabled). SeaweedFS runs in namespace `data` and exposes an S3-compatible API. A parallel Kubernetes Indexed Job in namespace `pipeline` reads `datasets/commands_pool.jsonl`, labels each command with a host-run Ollama `phi3.5`, and writes one raw shard object per worker. A cleaning step validates the replies and writes `datasets/v1/train.jsonl` with a SHA-256 checksum. Training distils the teacher replies into `EleutherAI/pythia-70m` and stores `model.tar.gz` and `model.sha256` in the `models` bucket. The FastAPI service runs in namespace `app`, downloads the model at start-up, and is exposed by a Traefik Ingress.

The main design choice is to keep weights out of the image and out of Git. The service image contains only Python and pinned libraries; `/readyz` stays 503 until the model has been downloaded, checksum-verified, extracted, and loaded. This makes cold start observable and makes a model update a change of `MODEL_URI`/`MODEL_VERSION`, not a rebuild. Ollama runs on the host rather than in the cluster because the teacher is only needed during Task 1 and host networking avoids a multi-gigabyte GPU image. The workers reach it through `host.k3d.internal:11434`. Every component is stateless except SeaweedFS, so Pod deletion and rolling updates are safe.

Version pins are recorded in the training configuration and the service, worker, and training dependency files. Cluster configuration is in `k8s/`; host-generated S3 credentials are kept in the git-ignored `.env.local`.

2. Task 1: object storage and parallel teacher labelling

2.1 Storage layout

SeaweedFS is deployed by `k8s/seaweedfs.yaml` into namespace `data`. Two buckets are created: `datasets` and `models`. `scripts/init_storage.py` uploads the provided `commands_pool.jsonl` to `datasets/commands_pool.jsonl`. `prompts/teacher_prompt.txt` asks for exactly the four keys and the eight allowed intents, gives three examples, and is fixed at `temperature=0`, `seed=42`, `top_k=1`.

2.2 Sharding and resumability

`worker/worker.py` assigns each input row to one shard by its line-number remainder and writes a JSONL shard containing the id, command, and raw teacher reply. The Indexed Kubernetes Job uses four completions and parallelism two. Each of the 3,000 input rows belongs to exactly one shard, so there are no gaps or repeats.

Resume is built in: before labelling, the worker reads its own existing shard object into a set of completed ids and skips them. Progress is written after every reply by default (atomic temp-file + rename), so an interrupted multi-hour run loses at most one command. The cleaner also refuses to proceed if ids are repeated across shards; `--allow-incomplete` is only for smoke runs.

2.3 Cleaning rules

`worker/clean.py` applies rejection precedence (a) -> (b) -> (c) -> (d) and counts each record exactly once:

- (a) `teacher_raw` is not valid JSON or is not an object;
- (b) keys are not exactly `intent`, `date`, `time`, `place`;
- (c) `intent` is not one of the eight allowed values;
- (d) a slot is neither null nor a string that appears in the command (case-insensitive, whitespace-normalised);
- duplicates are removed by normalised command text, keeping the smallest id.

Survivors are sorted by numeric id and written to `datasets/v1/train.jsonl`; the SHA-256 line is written to `datasets/v1/train.sha256`. The accounting invariant below is asserted in code:

`input = invalid_json + wrong_keys + bad_intent + bad_slot + duplicates + kept`

Category	Full run
Input records	3,000
Invalid JSON	49
Wrong keys	4
Bad intent	0
Bad slot	118
Duplicates	47
Kept	2,782

The invariant holds exactly: `3000 = 49 + 4 + 0 + 118 + 47 + 2782`. Coverage verification found 3,000 expected and present IDs, with no missing, repeated, or unexpected IDs. The cleaned dataset SHA-256 is `7e286136aaaaf4fd15e86ad4fe3c449b304182a69f7dd2664a0d729f13d33734`.

2.4 Sharding rationale

Four shards are the smallest count that gives two waves at `parallelism: 2`, so the cluster exercises both parallel workers and resume without creating idle pods. Sharding by line number, rather than by id, is exactly what the assignment specifies and is independent of the order of ids. Because every worker owns a disjoint residue class, the union is complete without a merge step. The cost is that a worker's commands are interleaved through the pool, but the worker sorts its own output by id, so each object is deterministic. Four workers also mean four objects, which is small enough to inspect by hand and large enough to show that the cleaning step really reads multiple shards. The local smoke run used the same code with four shards and one worker at a time, for 50 commands per shard (200 total), because a full CPU teacher run of 3,000 commands takes several hours; the Kubernetes Job is the intended full path and is resumable.

3. Task 2: training the student

3.1 Configuration

`training/config.yaml` records the six required items:

Item	Value
Random seed	42 (`torch.manual_seed`, NumPy, Python `random`; fixed in code)
Learning rate	5e-5
Batch size	4 (gradient accumulation 4, effective 16)
Epochs	2
Maximum sequence length	256
Library versions	Python 3.12, torch 2.4.1, transformers 4.44.2, tokenizers 0.19.1, safetensors 0.4.5, huggingface-hub 0.24.6, numpy 1.26.4, PyYAML 6.0.2, boto3 1.35.24

`training/config_v2.yaml` changes only the seed (43), epochs (3), version and loss file. `make train` reads only `training/config.yaml`; `make train SUBSET=50` reads the same file and caps the rows. `training/loss.csv` is written and flushed after every epoch.

3.2 Example format and loss

Each record becomes `Command: <text>\nJSON: <teacher reply>`. The prompt tokens are masked with `-100`, so the cross-entropy is computed only on the teacher reply tokens plus EOS; this matches "the model learns to produce the teacher's reply, given the command" and prevents the 70M model from spending capacity on reproducing the prompt. The model is `EleutherAI/pythia-70m`, loaded in float32, trained with AdamW and a linear warm-up schedule on CPU. The target is sequence-level: the exact teacher string, not logits.

Version	Records	Epoch losses	Seed	Wall time
v1	1,500	0.360529, 0.087982	42	403.2 s
v2	1,500	0.390328, 0.102742, 0.047950	43	787.8 s

3.3 Evaluation

Predictions are generated for the teacher (`phi3.5`, same prompt, temperature 0, seed 42), untrained Pythia-70M, and the two students. Raw outputs and score reports are in `results/`. The official `tools/score.py` computes JSON validity, intent accuracy, slot accuracies and full-frame accuracy.

Model	JSON validity	Intent	Date	Time	Place	Full frame
Teacher	99%	74%	74%	87%	79%	40%
Untrained Pythia	0%	0%	0%	0%	0%	0%
Student v1	99%	68%	75%	89%	75%	35%
Student v2	100%	71%	76%	89%	77%	38%

Version 2 improves JSON validity to 100%, intent accuracy to 71%, and full-frame accuracy to 38%. Its full-frame score is 95% of the teacher's 40%, while requiring one additional CPU training epoch.

3.4 Sequence-level versus logit-based distillation

Sequence-level distillation trains the student on the teacher's generated sequence, treating it as a hard target. The loss is the ordinary next-token cross-entropy on that sequence. Logit-based distillation trains the student to match the teacher's full next-token probability distribution at each position, typically with a temperature-softened KL divergence. Logit-based distillation is not possible here because the teacher is served by Ollama, which exposes only the generated text through `/api/generate`; it does not return per-token logits. Even if it did, the teacher and student use different tokenizers, so their vocabularies would need an alignment step. A limitation of the sequence-level approach is that it discards the teacher's uncertainty and cannot teach the student about near-miss alternatives; with only about 1,500 examples, the student can also memorise surface forms instead of generalising to unseen commands.

4. Task 3: inference service and container

The service is a FastAPI application on port 8000 with `POST /parse`, `GET /healthz` and `GET /readyz`. `/parse` returns only the four-key object and sets `X-Model-Version` and `X-Served-By` (the Pod name from a downward-API environment variable, with `HOSTNAME` and `socket.gethostname()` as fallbacks). The output validator checks exact keys, the eight intents, and string-or-null slots. If the model output is not valid, the service returns HTTP 500 with exactly `{"error":"invalid_model_output"}`; it never returns malformed JSON with 200. `FAULT_INJECT=1` replaces the generated text with `not json` before validation, which exercises that path.

At start-up the service downloads `s3://models/<version>/model.tar.gz` using boto3 against `S3_ENDPOINT`, verifies `model.sha256` if present, safely extracts the archive, and loads the model and tokenizer. `MODEL_URI`, `MODEL_VERSION`, `S3_ENDPOINT` and `MAX_NEW_TOKENS` come from the environment; in Kubernetes the two S3 credentials come only from the `model-store-credentials` Secret. The image is built by `service/Dockerfile` from `python:3.12-slim`, installs the CPU-only PyTorch wheel and pinned dependencies, and copies no weights. It runs as UID 10001 with a read-only root filesystem and a writable `/tmp` emptyDir for the model cache.

The Docker image built successfully at **1.63 GB** and ran in Kubernetes with the model fetched from SeaweedFS, not baked into the image. A local-process 50-request benchmark measured p50 **267.48 ms**, p95 **406.08 ms**, throughput **3.386 requests/s**, peak RSS **546.8 MiB**, and **50/50 successful requests** (0% errors; 14.767 s). The required separate Docker run constrained to 1 CPU and 1 GiB was not measured; local RSS is not a substitute for a container cgroup measurement. Generation is serialized by a lock.

5. Task 4: Kubernetes deployment, scaling and reliability

`k8s/app.yaml` creates a two-replica Deployment in `app` with CPU request 250m/limit 1, memory request 512Mi/limit 1Gi, liveness `/healthz` and readiness `/readyz`, `maxUnavailable: 0`, `maxSurge: 1`, a non-root read-only filesystem, and the Secret-backed S3 credentials. `k8s/ingress.yaml` routes `/parse`, `/healthz` and `/readyz` through Traefik; `k8s/hpa.yaml` scales 1-2 replicas at 70% CPU with a 60-second scale-down window.

The manifests parse successfully. Live validation was executed on 2026-11-08 with Docker 29.1.3, a two-node k3d cluster, and SeaweedFS. Both Kubernetes nodes became Ready, the v1 model archive was uploaded to the live `models` bucket, and two application replicas became Ready on separate nodes. Through a Traefik port-forward, `/healthz`, `/readyz`, and `/parse` each returned HTTP 200; ten additional inference requests also returned HTTP 200. The metrics API was available. A pod sample during the load period reported 992m CPU and 375 MiB memory; an earlier idle observation was about 395–398 MiB per pod. HPA scale-up and scale-down events were observed against the 70% CPU target.

The host-side k3d mappings on ports 8080 and 8333 returned empty responses, so the live checks used supported `kubectl port-forward` paths. The following additional live measurements were captured: the v1-only rolling update and rollback completed, pod recovery took 12 seconds, three cold starts took 11, 11, and 10 seconds, and a short Ingress load test completed 76 requests with zero failures (p50 6.0 seconds, p95 6.6 seconds, 1.297 requests/second). HPA scale-down to one replica and subsequent scale-up were also observed. The short CPU-bound test sent all successful requests to one replica, so balanced per-replica request share remains unmeasured. The v2 artifact upload remains blocked by SeaweedFS reporting `no free volumes left`; the v1-only rollout validates Kubernetes behavior but not a real v1-to-v2 model change.

Local limitation: the cluster is two k3d nodes on one laptop and shares one CPU and one disk. Scaling to two replicas demonstrates the mechanism but does not provide machine-level fault tolerance or a realistic multi-node network. A real limitation of the HPA is that CPU utilisation includes model download and start-up work; a pod can be scaled before it is Ready, and the readiness probe is what protects the Service from traffic. During a rolling update the old pod remains Ready until the new pod downloads and loads the model, so the update can take as long as a cold start.

6. Task 5: subnetting and NACL

Requirements are allocated largest first, each immediately after the previous block, with the smallest prefix that fits:

Subnet	Network	Prefix	Usable host range	Broadcast	Usable
app	10.20.0.0	/22	10.20.0.1- 10.20.3.254	10.20.3.255	1022
public	10.20.4.0	/24	10.20.4.1- 10.20.4.254	10.20.4.255	254
data	10.20.5.0	/25	10.20.5.1- 10.20.5.126	10.20.5.127	126
management	10.20.5.128	/27	10.20.5.129- 10.20.5.158	10.20.5.159	30

The app subnet needs 1,000 hosts, so /22 (1,022 usable) is the smallest fit; public needs 250, so /24 (254); data needs 100, so /25 (126); management needs 20, so /27 (30). The total consumed is 10.20.0.0/22, 10.20.4.0/24 and the two halves of 10.20.5.0/24, leaving space for growth.

`network/nacl.json` is the public-subnet NACL. Inbound rule 100 denies 198.51.100.0/24 for all protocols and ports before the web allows; rule 110 allows TCP 80, rule 120 allows TCP 443, rule 130 allows TCP 22 only from 10.20.5.128/27, and rule 140 allows return traffic from 10.20.0.0/22 on ephemeral ports 1024-65535. Outbound rule 100 allows TCP 8000 to 10.20.0.0/22, and rule 110 allows replies to clients on ports 1024-65535; there is an implicit deny for everything else. The official simulator reports **6 / 6 = 100.0%**.

A NACL is stateless: it does not remember that an inbound connection created a flow, so the response packets are evaluated as new packets and need their own rules in both directions. Rule order matters because NACL rules are evaluated from the lowest number and the first match wins; the deny for 198.51.100.0/24 must be numbered below the TCP 80/443 allows, otherwise those allow rules would match first and the abusive range would be allowed.

7. Task 6: reproducibility, testing and CI/CD

`make up` creates the cluster, secrets, SeaweedFS, buckets, images, uploads the model and applies the Deployment/Service/Ingress/HPA; `make down` deletes the cluster. From a fresh clone the whole system is two commands: `make install-ml` and `make up`, or just `make up` because the target bootstraps the virtual environment if it is missing. `make up` does not commit credentials: `scripts/create_secrets.sh` generates them, writes them to the git-ignored `.env.local` with mode 0600, and creates the Kubernetes Secrets.

The test suite has 49 tests covering the cleaner's rejection precedence and duplicate rule, accounting invariant, sharding and resume, subnet arithmetic, the official NACL simulator, service API and exact error body, readiness before load, output

validation, storage backends, and cost arithmetic. `ruff check .` passes, `pytest -q` reports `49 passed`, and `gitleaks detect --no-git` reports `no leaks found` (evidence in `results/`).

The CI workflow in `.github/workflows/ci.yml` runs in the required order: `lint` (ruff) -> `unit-tests` (pytest) -> `model-evaluation` (downloads `model.tar.gz` and `model.sha256` from the `final` GitHub Release, verifies the checksum, runs `training/predict.py`, then `tools/score.py`) -> `quality-gate` (`--min-validity 95`). A separate `secret-scan` job runs gitleaks. A manual `workflow_dispatch` input `bad_model=true` runs the untrained Pythia-70M through the same gate and is expected to fail.

The CI run history for this repository, including the normal run and the `bad_model=true` quality-gate demonstration, is in its GitHub Actions tab.

`AI_USE.md` lists the AI tools and the checks applied; every generated file was reviewed, linted, and tested, and the measured results come from commands in `results/`, not from the assistant.

8. Task 7: performance, cost and review

The local 50-request benchmark measured p50 **267.48 ms**, p95 **406.08 ms**, throughput **3.386 requests/s**, 50 total requests, and 0% errors. Ingress Locust results are 76 requests, 0 failures, p50 **6.0 s**, p95 **6.6 s**, and **1.297 requests/s**. The cold-start median is **11 s**. All results are in `results/`; the short CPU-bound run did not demonstrate balanced replica use.

8.1 Cost

A replica is billed for its CPU limit and memory limit for every hour it runs, so with two replicas at 1 vCPU and 1 GiB each:

```
per replica per hour = 1 * 3.50 + 1 * 0.45 = 3.95 INR
all replicas per hour = 2 * 3.95 = 7.90 INR
requests per hour = measured requests/second * 3600
cost per 1,000 = (7.90 / requests per hour) * 1000
```

Using the short Ingress run (1.297 requests/s, 76 responses, 0 HTTP failures), the derived cost is approximately **1.69 INR per 1,000 successful HTTP responses**, assuming two replicas billed at their limits for the entire hour. This is an estimate from a short run, not a sustained capacity or semantic correctness measurement; Locust counted HTTP success, not frame accuracy.

`scripts/cost.py` performs exactly this calculation from `results/loadtest_stats.csv` and `data/rate_card.json`, and writes `results/cost.json`.

8.2 SLO

SLO: **99.5% of `/parse` requests succeed (HTTP 200) over one load-test window**. For the cluster window, N=76 and the error budget is `floor(0.005 * 76) = 0` failures. Actual failures were 0, so that measured window met the SLO. This does not certify sustained production availability.

8.3 Review

Task	Measured result	Limitation	Improvement
1. Storage and labelling	3,000 labels; 2,782 kept; accounting/coverage pass	SeaweedFS exhausted volumes during v2 upload	Restore capacity and verify v2 upload
2. Student training	1,500 examples/version; v2 JSON validity 100%, full-frame 38% vs teacher 40%	Only 100 validation examples	Expand held-out validation set
3. Service and container	1.63 GB image; Kubernetes inference and local 50/50 pass	1 CPU/1 GiB container benchmark not measured	Repeat constrained benchmark with cgroup metrics
4. Kubernetes	Ingress 76 requests/0 failures; recovery 12 s; cold-start median 11 s	Short run used one replica; v1-only update	Longer test and true v1-to-v2 rollout

Task	Measured result	Limitation	Improvement
5. Networking	NACL simulator 6/6 correct	Simulated packets, not a cloud deployment	Validate rules in the target cloud
6. Testing and CI	49 tests pass; CI and expected quality-gate failure recorded	No dedicated multi-node CI runtime	Add a disposable cluster integration job
7. Performance and cost	p95 6.6 s; 1.297 RPS; estimated 1.69 INR/1,000 HTTP 200	Short-run estimate, not semantic success cost	Measure sustained throughput and correct frames

9. Limitations and improvements

The main operational limitation is SeaweedFS volume exhaustion: the service reported `no free volumes left` while uploading the v2 archive. The live deployment and Kubernetes behavior were therefore validated with v1, while a real model artifact switch remains unverified. The short Locust run also sent all recorded successful requests to one replica, so it does not demonstrate balanced request distribution. The image was built and run, but the separate 1 CPU / 1 GiB Docker benchmark was not measured.

Beyond the environment, a 70M-parameter student will not match a 3.8B teacher; despite full 1,500-record training, the 100-example validation set is small and the student's full-frame score remains below the teacher. The HPA reacts to CPU rather than to queue depth, the service serialises generation under a lock, and the k3d cluster is two nodes on one machine. Better choices would be a queue-depth or latency-based autoscaler, a batched/streaming inference server, and a multi-machine cluster to test real node failure. On the data side, more teacher examples, stricter slot grounding, and a held-out split would reduce overfitting. On the security side, the S3 credentials are generated and Secret-backed, the image is non-root and read-only, and the NACL is default deny; a network policy restricting the app namespace to SeaweedFS and the Ingress would add defence in depth.